MILITARY SPECIFICATION

RECEIVER-COMPARATOR R-1407( )/URQ

1. SCOPE

1.1 This specification covers requirements for a transistorized Very Low Frequency (VLF) Receiver and Phase Comparator, to be used for comparison of a local frequency standard against standard VLF transmissions. The receiver covers the tuning range of 9 kilocycles (kc.) to 30 kc., in 100 cycle per second (c.p.s.) increments, and 60 kc.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-C-175 - Cable Assembly, Power, Electrical (3 wire, 3 prong, Grounding Plug Connector, for 125-Volt Equipment).

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore; General Specification.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-189 - Racks, Electrical Equipment, 19-Inch and Associated Panels.
MIL-STD-449 - Radio Frequency Spectrum Characteristics, Measurement of.

FSC 5820

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:54Z
Check the source to verify that this is the current version before use.

DRAWINGS

BUREAU OF SHIPS
REA49028 - UG625B/U - Connector, Receptacle Series BNC.

PUBLICATION

BUREAU OF SHIPS
NAVSHIPS 93820 - Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- The equipment shall be in accordance with the requirements of MIL-E-16400, except as otherwise specified herein.

3.1.1 Qualification.- The equipment furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2). One receiver shall be submitted for qualification testing.

3.1.2 Reliability.- The predicted reliability for the equipment shall be at least 5,000 mean-time between failures (MTBF). MTBF of the equipment shall be predicted using Method D of NAVSHIPS 93820. For test purposes, 300 hours MTBF will be demonstrated. The equipment shall meet the requirements of level 3 of MIL-R-22732.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the current version before use.

3.2 Mechanical design.- The equipment shall be capable of 19-inch rack mounting and separate shelf mounting in accordance with MIL-STD-189. Drawer slides shall be provided for withdrawing the equipment from the enclosure or rack. Latches shall be provided to lock the drawer slides in the external position for servicing. The equipment when withdrawn, shall be capable of being rotated and locked in 3 positions 90 degrees apart for servicing top or bottom of the chassis.

3.2.1 Shock.- The equipment shall meet the requirements for Grade A, Class 1, lightweight, Type A of MIL-S-901.

3.2.2 Degree of enclosure.- The enclosure shall be dripproof in accordance MIL-STD-108.

3.2.3 Size and form.- Maximum dimensions of the equipment shall be as follows:

Height - 7 inches (including front panel).
Width - per MIL-STD-189.
Depth - 18 inches.

3.2.4 Weight.- The maximum weight of the equipment shall be 50 pounds.

3.2.5 Temperature class.- The equipment shall meet class 4 temperature requirements of MIL-E-16400.

3.3 Electrical design.- The equipment shall be designed to provide the following performance characteristics over the environmental conditions and voltage variations specified by MIL-E-16400. The equipment shall be capable of continuous operation under these conditions.

3.3.1 Frequency range.- Minimum frequency range shall be 9 kc to 30 kc selectable in 100 cps increments, with no hiatus in this range; also 60 kc.

3.3.2 Frequency selection.- The tuning mechanism shall provide positive lock-in on the desired frequency. The required performance shall be obtained on all channels without further adjustments (except gain control), switching, or plug-in devices. A read-out, directly indicating the set frequency shall be provided.

3.3.3 Receiver sensitivity.- The equipment shall be capable of tracking a signal with an off-set of 1 pp 10^8 with an input signal level of 0.01 microvolt (uv) across 50 ohms, and with a signal-to-noise ratio of -50 decibels (db) (Gaussion noise measured in 1 kc bandwidth).

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the current version before use.

3.3.3.1 Sensitivity range.- Total signal operating range shall be 120 db minimum, of which at least 40 db shall be inherently or automatically compensated, with maximum phase shift of ± 0.25 microseconds (μsec).

3.3.4 Receiver response.- Response of the receiver shall be based on a desired signal input level of 1 μv with all receiver controls optimized for reception of the desired signal before application of the interfering signal.

3.3.4.1 Adjacent channel rejection.- Phase tracking of a desired signal shall not be affected by a signal ± 100 c.p.s. from and 50 db stronger than the desired signal.

3.3.4.2 Undersired response.- All undesired signals from 100 c.p.s. to 1 mc beginning 200 c.p.s. on either side of the set frequency, shall be rejected at least 60 db. It is desirable that the image frequency fall outside the tunable range.

3.3.4.3 Desensitization.- An undesired signal 200 c.p.s. from the desired channel and 90 db stronger than the desired channel shall not affect gain of the receiver more than 3 db.

3.3.5 Local signal radiation.- Any internally generated signal present at the 50 ohm antenna input shall not exceed -167 dbm.

3.3.6 Receiver bandwidths.- The following 3 db bandwidths shall be provided:

(a) Intermediate Frequency (IF) - 200 c.p.s.
(b) 1 kc Envelope Output - 200 c.p.s.
(c) Servo Predetection - 40 c.p.s. minimum
(d) Servo Postdetection - 0.005 c.p.s. nominal.

3.3.7 Phase tracking servo.- A servo shall be incorporated to maintain continuous tracking of the received signal. Phase error between the received signal and a signal synthesized from the reference standard input shall be detected; this error signal shall drive the servo. Servo tracking shall be indicated on the chart recorder, and on a re-settable digital read-out indicating time difference and readable from .1 microseconds to 9999.9. The servo deadband shall be ± .1 microsecond, maximum. The servo shall be capable of tracking a signal changing in phase at a rate of 0.1 microsecond per second, or slower, with a received signal input of .01 microvolt and signal to noise ratio of 1:1. Maximum tracking error under these conditions shall be 5 microseconds. For increasing delay (low frequency) of the reference input, the servo phase indication shall increase, and the recorder meter shall swing to the right.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the current version before use.

3.3.7.1 Servo slewing.- A slewing control shall be provided to manually advance and retard the servo at a minimum rate of 0.2 microseconds per second.

3.3.7.2 Servo disable.- The servo shall be automatically disabled whenever the received signal level drops below that required for reliable phase tracking. The servo shall reactive automatically at an input level of .01 μv. A manual lock-out switch shall be provided to by-pass this automatic feature.

3.3.8 Chart recorder.- A built-in inkless chart recorder shall be provided to present a continuous recording of phase relationship between the received signal and the signal synthesized from the local source (frequency standard). The recorder shall be mounted so that it can be observed from the front of the equipment. Access shall be provided for writing on the chart paper. The paper shall be capable of being unrolled for viewing the previous 72-hour period. A convenient means for re-rolling the chart shall be provided. The recorder mechanism shall be essentially silent in operation. A switch shall be provided to disable the chart feed motor without interferring with operation of the remainder of the equipment. The chart scale shall have 10 major divisions and 50 minor divisions. Full scale shall be 10 or 100 microseconds, switch selectable. Time of day increments on the chart shall be 15 minutes (or smaller). Chart capacity shall be at least 30 days for continuous operation. Chart rate shall be 1-inch per hour.

3.3.9 Frequency synthesizer.- The equipment shall incorporate a frequency synthesizer to generate a local oscillator signal in 100 c.p.s. increments, phase coherent with an external source of either 1 mc. or 100 kc. Switching away and returning to any set frequency shall result in a phase shift of no more than 0.05 μsec.

3.3.10 Calibration accuracy.- The calibration accuracy (relative to a received carrier) shall be better than $\pm 1 \times 10^{-11}$ on a 24 hour basis over the specified temperature range and input AC line voltage variation. The total receiver phase shift from all causes shall be less than ± 0.5 microseconds for a temperature range of 0°C to +50°C.

3.3.11 Inputs.- Connectors shall be located on rear of the equipment.

3.3.11.1 Antenna.- The equipment shall be capable of operation with a loop antenna or a long wire antenna. Input impedance shall be 50 ohms (nominal) for the loop antenna and 10,000 ohms (minimum) for the long wire antenna.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T19:54Z
Check the source to verify that this is the currentversion before use.

3.3.11.2 Reference input.- The equipment shall accept an external reference at either 100 kc, or 1 mc; sine wave between 0.5 and 5.0 volts rms or square wave 0.5 to 5.0 volts peak to peak.

3.3.12 Outputs.- Connectors shall be located at rear of the equipment.

3.3.12.1 Recorder.- Relative phase information shall be available for connection to an external recorder. Level shall be 1 milliamp (ma) for 10 or 100 μsec phase difference, as determined by the scale switch. A level adjustment shall be provided for calibrating an external recorder with any impedance between 1,000 and 6,000 ohms.

3.3.12.2 Coherent outputs.- Square wave signals at 100 kc, 10 kc, 1 kc and 100 c.p.s. shall be provided which are phase coherent to the received signal; minimum level shall be 1 volt peak to peak into 600 ohms.

3.3.12.3 1 kc.- A 1 kc sine wave output shall be provided. This signal shall carry the same phase information as the input carrier, and have a nominal level of 1 volt rms into 600 ohms.

3.3.12.4 Audio.- An audio section with speaker and volume control shall provide for aural monitoring of the received signal.

3.3.12.5 Reference.- An output of 100 kc, square, pulse, or sine, phase coherent to the reference input shall be provided. Minimum level shall be 1 volt rms into 600 ohms.

3.3.13 Front panel meter.- This meter shall be provided to monitor various equipment functions, switch selectable, as follows:

3.3.13.1 Relative carrier level.- The meter shall be marked to show normal operating range as follows:

(a) Servo phase detector error voltage; zero-center.
(b) Power supply voltage(s).
(c) Frequency synthesizer output level.
(d) External reference input level.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the currentversion before use.

3.3.14 Indicators.- Indicators shall be as follows:

(a) Receiver frequency.
(b) A.C. power-on.
(c) Insufficient incoming signal-on.
(d) Power/external standard interrupt-on. This indicator shall lock on when an interruption occurs in either receiver power or the signal from the external reference. The interrupt re-set (3.3.15(l) shall extinguish this indicator.
(e) Blown fuse indicators.

3.3.15 Controls.- All controls shall be continually required, except (f), (g), (h) and (k), which shall be occasionally required. Locking devices shall be provided as necessary to prevent inadvertent movement of controls due to shock and vibration. Illumination of controls is not required:

(a) Receiver gain.
(b) Receiver frequency selection.
(c) Servo OFF/TRACK/CUT-OUT BY-PASS.
(d) Servo + SLEW/-SLEW (spring return to center).
(e) Meter function selector.
(f) Full-scale phase error selector.
(g) Recorder zero and full-scale adjustment.
(h) Recorder motor ON-OFF.
(i) Accumulated time re-set.
(j) Audio level.
(k) External recorder full scale adjust.
(l) Interrupt re-set.
(m) POWER-ON/OFF

3.3.16 Power requirements.- The equipment, excluding the chart drive motor, shall be powered by 115 volts, 50 to 400 c.p.s., single phase or 26 volts ± 4 direct current (d.c.), negative ground. A detachable line cord and plug conforming to type II of J-C-175 shall be supplied for connection to the 115 volt source, attached to the rear of the equipment through chassis receptacle type MS3102E-14S-7P and mating cable connector type MS3106E-14S-7S in accordance with MIL-C-5015. Chassis connector type MS3102E-12S-3P shall be located on the rear of the equipment for connection to the d.c. source. Mating cable connector type MS3106-12S-3S shall be supplied. The equipment shall be capable of operating from the d.c. source in event of a.c. power failure with no interruption of any function except the chart recorder. The chart drive motor shall operate on 115 volts, 60 cycle, single phase. For power frequencies other than 60 cycle, proper chart rate will not be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:54Z
Check the source to verify that this is the current version before use.

3.3.17 Time meters are not required.

3.3.18 Signal connectors.- Except as otherwise specified, all signal connections external to the equipment shall be made through BNC connectors, type UG-625B/U of REA49028.

3.3.19 Instruction plates.- Instruction plates is used shall be as specified in MIL-E-16400.

3.3.20 Commutator operation.- A connector shall be provided to supply power and time base signal to an external commutator and to carry blanking signals from the cummutator to the receiver IF circuit. The receiver shall be blanked and the servo disabled during blank command from the commutator. The receiver shall track the received signal during the unblank interval. The receiver shall be unblanked with zero volts applied to the gate input, and blanked with plus 1 volt applied. Minimum gate input impedance shall be 1,000 ohms. No damage shall occur with levels up to ± 20 v peak applied. The receiver shall be capable of normal operation on non-sequenced stations, when the commutator is disconnected or turned off.

3.3.21 Receiver delay.- Receiver delay shall be determined and the nominal value shall be specified in the technical manual. Delay shall not vary more than ± 3 μsec from the nominal value for any received frequency.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests 1/.- Qualifaction tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.4.

4.3 Test procedure approval.- Procedures for all required inspections shall be submitted to the bureau or agency concerned for approval at least 30 days before tests are scheduled to begin. Procedures shall include descriptions of test equipment.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the currentversion before use.

4.4 Qualification inspection.- Qualification inspection shall consist of the tests shown in table I.

Table I - Qualification inspection.

Inspection	Reference
Group I	
Surface examination	MIL-E-16400
Operating test	MIL-E-16400
Weights and dimensions	MIL-E-16400
Heat	MIL-E-16400
Enclosure	MIL-E-16400
Power	MIL-E-16400
Radio interference and undesired radiation	MIL-E-16400
Supply line voltage and frequency	MIL-E-16400
Group II	
Frequency range and selection	4.6.2
Sensitivity	4.6.3
Sensitivity range	4.6.4
Receiver response	4.6.5
Adjacent channel rejection	4.6.5.1
Undesired response	4.6.5.2
Desensitization	4.6.5.3
Local signal radiation	4.6.6
Receiver bandwidths	4.6.7
Servo deadband	4.6.8
Servo rate	4.6.9
Servo slewing	4.6.10
Servo disable	4.6.11
Chart recorder	4.6.12
Synthesizer phase shift	4.6.13
Calibration accuracy	4.6.14
Reference input	4.6.15
Recorder output	4.6.16
Coherent outputs	4.6.17
Audio output	4.6.18
Panel meter	4.6.19
Controls and indicators	4.6.20
Commutator operation	4.6.21

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the currentversion before use.

GROUP III

Temperature	MIL-E-16400
Humidity	MIL-E-16400
Shock, vibration, inclination	MIL-E-16400 and 4.6.5.23
Salt spray	MIL-E-16400
Reliability	4.6.22

4.5 Quality conformance inspection.- Quality conformance inspection shall include production inspection of 4.5.1 and production control inspection of 4.5.2.

4.5.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery and shall comprise such examination and tests as will prove the workmanship and reveal possible omissions and errors in the production process. Production inspection will be in accordance with MIL-E-16400.

4.5.2 Production control inspection.- Production control inspection shall be conducted on production equipments (see 4.5.1). Inspection shall be conducted to detect any deterioration in the design by wear of such items as dies, molds, jigs, fixtures, and to detect deviations in the material or part processing, selection, or assembly, and to detect any deterioration in the performance of the equipment not discernible under the testing of 4.5.1. Sample units shall be selected in accordance with table 11.

Table II - Sampling for production control inspection

Cumulative number of equipments produced	Cumulative number of equipments to be tested after 0, 1, 2, 3 equipments have failed in any test.			
	0	1	2	3
1 to 3	2	3	--	--
4 to 6	3	5	--	--
7 to 10	5	8	--	--
11 to 15	6	10	--	--
16 to 25	8	12	16	--
26 to 40	10	15	20	25
41 to 65	12	18	24	30
66 to 95	14	20	28	35
96 to 145	17	25	34	42
146 to 200	20	30	40	50
201 to 290	24	35	50	55
291 to 440	30	42	60	70

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:54Z
Check the source to verify that this is the current version before use.

4.5.2.1 Each of the units selected in 4.5.2 shall be subject to the tests Groups I and II shown in table I, except radio interference and undesired radiation; a unit shall be rejected if if fails in any one of them. The number of sample equipments which shall be inspected shall be increased for each failure, as shown in Table III. For example: Assume that 40 equipments have been produced, 10 of which have been subjected to the production control inspection specified herein, and all of these equipments have passed every test. From the next 25 produced, two shall be tested, provided both of them pass every test. But if one equipment fails, six additional equipments shall be selected from the increment of 25. If two of these fail in any test, another six shall be tested bringing the total number tested to 24. Thereafter, sampling shall be continued in accordance with the column headed 2 (failures).

4.5.2.2 After any change has been made (see 4.5.3), sampling shall be started with the first entry in table III and progress downward as for a new equipment design.

4.5.3 If at any time the number of equipments failing in any test exceeds the total number allowed, the contractor will be required to make such changes as may be necessary to meet the requirements of this specification. If the remedial action required to correct the equipment performance is a change of parts(s), circuit lay-out, or the mechanical equivalent, such action shall be reported immediately to the bureau or agency concerned. It shall then be determined by the bureau or agency whether the correction shall be made in the equipments already delivered under the contract.

4.5.4 Environmental tests.- Group III tests shown in table I and interference and undesired radiation shall be performed on the first of each lot of 100 equipments on a production order. However, shipment of lots shall not be withheld pending results of the environmental tests.

4.6 Test procedures.-

4.6.1 Test conditions.-

4.6.1.1 Dummy antenna.- For all operating tests the signal source shall be connected to the antenna terminals of the receiver through a matching network, which, in combination with the signal source presents a 50 ohm or 10,000 ohm impedance (depending on which antenna terminal is being fed) to the receiver input. The signal source shall be a frequency synthesizer driven from a frequency standard having an average drift of more than 1.5 pp 10^{10} per day. Each antenna input shall be used for one or more of the following tests.

4.6.1.2 Standard response.- For purpose of these tests, standard response shall be defined as faithful phase tracking by the receiver servo of an incoming signal which is offset from the reference signal by 1 pp 10^{8}.

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:54Z
Check the source to verify that this is the current version before use.

4.6.1.3 Standard frequencies.- For purpose of these tests, standard frequencies are defined as 10.1 kc., 19.5 kc., 28.9 kc., and 60.0 kc. Unless otherwise specified, the test shall be performed at each frequency.

4.6.2 Frequency range and selection.- Standard response shall be obtained at 9.0 kc., 18.8 kc., 30.0 kc., and 60 kc. with 1 μv signal level. The tuning mechanism shall be operated over sufficient points to determine compliance with all requirements of 3.3.1 and 3.3.2.

4.6.3 Sensitivity.- This test shall be performed at standard frequencies which are offset from nominal by 1 pp 10^8 with respect to the reference signal. The signal input level shall be no greater than 0.01 μv with a signal-to-noise ratio of -50 db (Gaussian noise measured in 1 kc bandwidth). The receiver shall faithfully track the standard frequencies. The equipment shall also be capable of tracking 1 pp 10^8 offset with input signal level of 10 mv.

4.6.4 Sensitivity range.- This test shall be performed at 10.1 and 28.9 kc. Signal input level shall be .01 μv initially. Standard response shall be maintained without adjusting the receiver gain control, as signal level is increased to 1.0 μv. The test shall be repeated with receiver gain control set at maximum signal position, and again at approximately mid-range, with the input signal level varied over a 40 db range. Phase shift due to signal level change shall be no more than ± 0.25 μsec.

4.6.5 Receiver response.- This test shall be performed at two of the standard frequencies (offset less than 1 pp 10^{10}). The standard frequency shall be designated the desired signal. The desired signal level shall be 1 μv rms. All receiver controls shall be optimized for reception of this signal.

4.6.5.1 Adjacent channel rejection.- An undesired signal 100 c.p.s. from the desired signal and 50 db greater than the desired signal shall be inserted into the matching network simultaneously with the desired signal. Faithful phase tracking of the desired signal shall be observed. A step phase shift shall be made in the desired signal to prove that the receiver is tracking the proper signal.

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:54Z
Check the source to verify that this is the current version before use.

4.6.5.2 Undesired response.- Receiver response shall be measured at the coherent 1 kc sine output. Tests shall be made with the receiver tuned to 20.0 kc. Tests shall be performed by feeding the antenna input with signals (one at a time) of 10 mv at the following frequencies:

(a) f_{IF} (all IF's)

(b) $f_r \pm 2f_{IF}$ (all IF's) (image)

(c) $n(f_r \pm f_{IF}) \pm f_{IF}$ for n = 2.3 (first IF) (conversion)

(d) f_r/n for n = 2.3 (subharmonic)

(e) 100 c.p.s. 1 kc. 10 kc. 100 kc. 1 mc (decade)

where f_{IF} = intermediate frequency

f_r = received (tuned) frequency

n = an integer

Requirements of 3.3.4.2 apply over the entire frequency range stated therein. The above test may be performed at other critical frequencies to determine compliance.

4.6.5.3 Desensitization.- This test shall be performed with the receiver tuned to standard frequencies at 1 μv level. Receiver gain shall be determined. Undesired signals at 10.3, 19.7, 28.7 and 58.8 kc, respectively, at 31.6 mv level shall then be inserted. Receiver gain shall drop no more than 3 db.

4.6.6 Local signal radiation.- Measurements shall be made to determine conformance with 3.3.5. Procedures for the oscillator radiation test specified in MIL-STD-449 shall be used as a guide.

4.6.7 Receiver bandwidths.- Measurement of bandwidths shall be made at appropriate points within the receiver to determine compliance with 3.3.6.

4.6.8 Servo deadband.- The servo is slewed off in a positive direction by at least 1 μseconds and allowed to return, taking care to allow enough time for a true null. A reading is taken of the time difference register. The servo is then slewed off in a negative direction by at least 1 μsecond and allowed to fully null. A reading is taken of the time difference register. The difference of the two readings is the servo deadband and shall be less than 0.2 μseconds (± 0.1 μs)

Source: https://assist.dla.mil -- Downloaded: 20261O-19T19:54Z
Check the source to verify that this is the current version before use.

4.6.9 Sevo rate.- Test shall be performed at one of the standard frequencies. Signal input level shall be .01 μv, with 1:1 signal-to-noise ratio. Faithful tracking without overshoot shall be obtained as the reference frequency is offset from zero to 1 pp 10^7 in either direction with respect to the received signal. Phase indication shall be in the direction required by 3.3.7.

4.6.10 Servo slewing.- Conformance with 3.3.7.1 shall be determined.

4.6.11 Servo disable.- Conformance with 3.3.7.2 at standard frequencies shall be determined.

4.6.12 Chart recorder.- The equipment shall be examined to determine compliance with 3.3.8.

4.6.13 Synthesizer phase shift.- The equipment shall be tuned to standard frequencies. Input signal level shall be 1 μv. The equipment shall be retuned to any convenient signal unrelated to the test frequency, and at least 300 cycles removed, for a sufficient time to ensure that the receiver is tracking the new frequency. Tuning shall then be returned to the standard frequency. Phase relationship shall return to within ± 0.05 μseconds of the reading prior to first retuning.

4.6.14 Calibration accuracy.- This test may be combined with temperature tests and the supply line voltage tests. Reference input and antenna input shall be energized with signals derived from the same synthesizer source, so that the phase relationship between the two inputs can be determined. Measurements shall be made to determine compliance with 3.3.10.

4.6.15 Reference input.- The equipment shall be operated at one of the standard frequencies. The following combination shall be used to drive the reference input to determine compliance with 3.3.11.2:

Sine	100 kc	5.0v rms
Square	1 mc	.5 v peak to peak

Input impedance, measured at 1 mc shall be 1,000 ohms or greater.

4.6.16 Recorder output.- An external meter or recorder with appropriate parameters shall be connected to the recorder output connector to determine compliance with 3.3.12.1. The equipment shall be operated at one of the standard frequencies.

4.6.17 Coherent outputs.- The outputs required by 3.3.12.2, 3.3.12.3 and 3.3.12.5 shall be monitored while operating the receiver at one standard frequency, and varying the phase of the input through at least 360° (one cycle).

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:54Z
Check the source to verify that this is the current version before use.

4.6.18 Audio output.- The equipment shall be operated at one standard frequency, at 0.1 μv input level to determine compliance with 3.3.12.4.

4.6.19 Panel meter.- The meter and it's associated switch shall be operated to determine conformance with 3.3.13 and 3.3.13.1.

4.6.20 Controls and indicators.- Controls and indicators shall be operated as necessary to determine conformance to 3.3.14 and 3.3.15.

4.6.21 Commutator operation.- The gate input level (dc) required to blank the receiver shall be determined for compliance with 3.3.20. Plus and minus 20 v dc shall be applied to determine that no damage occurs.

4.6.22 Reliability testing.- Tests shall be performed as specified in MIL-R-22732 for test level number 3. Minimum reliability level shall be 300 hours. If an accept decision cannot be made before 720 hours test time on an equipment, the test shall be terminated and test data submitted to the Bureau for decision to accept or reject. The equipment shall be operated with 19.9kc input at .1μv level; standard response shall be obtained throughout this test. Any break in tracking shall constitute a failure.

4.6.23 Shock test shall be conducted in accordance with procedures in MIL-S-901 for Grade A, Class 1, lightweight, Type A equipment.

4.6.24 Humidity.- Performance during humidity test shall be the same as specified in 4.6.22.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for a storage of shipboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during handling and direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:54Z
Check the source to verify that this is the current version before use.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, and contractor's name and destination.

5.1.2 Shipboard repair parts.- Repair parts shall be preserved and packaged level A, packed level C and marked levels A or C in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment requirements.-

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing, and marking.- The equipment, accessories, repair parts, and technical publications shall be preserved and packaged level A or C, packed level A or B, marked level A or C, in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of preservation and packaging, and the level of packing when required (see 5.2.1).

6.2 With respect to products requiring qualification, awards will be made only for such products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24195, whether such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:54Z
Check the source to verify that this is the current version before use.

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Preparing activity:
Navy - SH
(Project 5820-N472Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE D14000

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:54Z
Check the source to verify that this is the current version before use.

Defense

FOLD

DEPARTMENT OF THE NAVY
BUREAU OF SHIPS
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

CHIEF BUREAU OF SHIPS
SPECIFICATIONS AND STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:54Z
Check the source to verify that this is the current version before use.